

BITS PILANI, DUBAI CAMPUS
Dubai International Academic City, Dubai.
Third Year - II Semester 2016-17
Design and Analysis of Algorithms (CS F364)
Test -2 (Open Book)

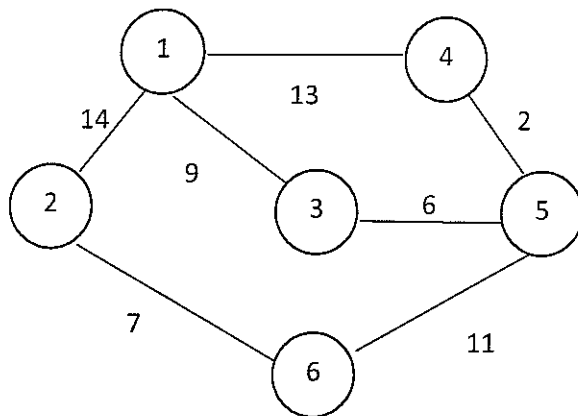
Date: 13.04.2017

Max Marks: 40

Time: 8.30 to 9.20 am, 50 minutes

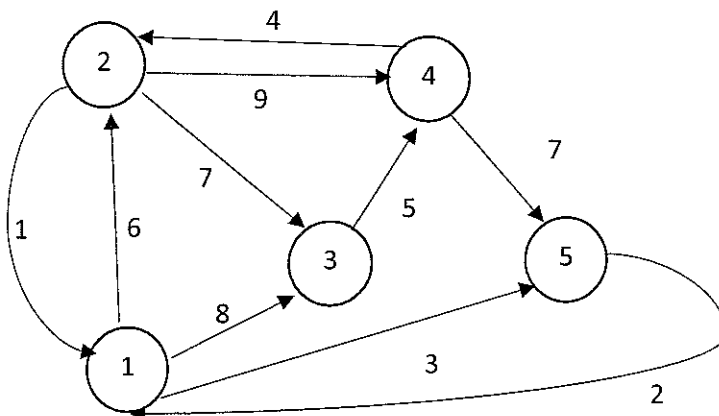
Weightage: 20%

1. For given graph, determine minimum spanning tree using Prim's algorithm. Provide the adjacency matrix representation for graph. Also show the content of array 't' and 'near' of the Prim's algorithm for each step. What is the cost of minimum spanning tree?



[10]

2. Compute the cost matrix for all pair shortest paths for given graph.

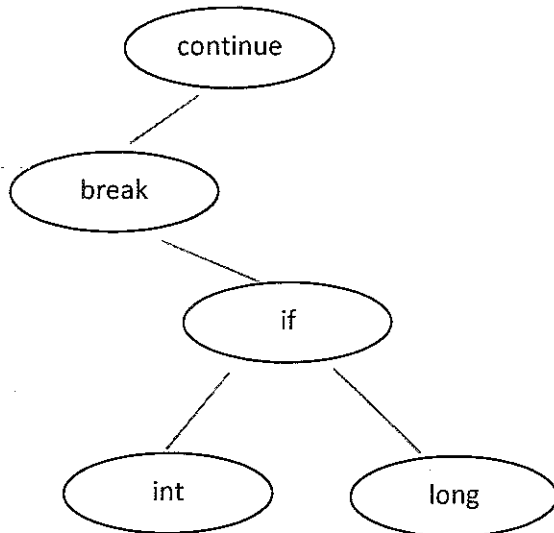


Clearly show A^0 , A^1 , A^2 , A^3 , A^4 and A^5 matrices.

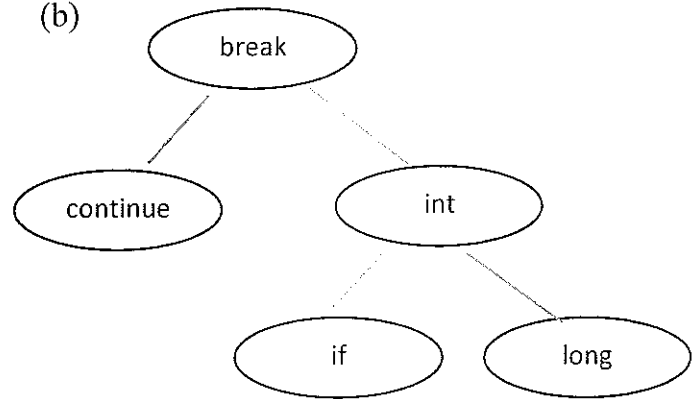
[10]

3. For given optimal search trees, determine total search cost for tree (a) and (b). Symbols are {if, continue, break, long, int}. Assume equal probability for each internal and external nodes.

(a)



(b)



[8]

4. Consider an array of characters. Write non deterministic algorithm for sorting character in decreasing order.

[6]

5. Consider chromatic number decision problem. We have given following CNF with 3 literals. Construct the graph on given CNF based on satisfiability with at most 3 literals per clause reduces to chromatic number decision problem. Clearly write vertices set V, edge set E and draw the resulting graph.

CNF is $(x_1 \wedge x_2 \wedge \bar{x}_3) \vee (\bar{x}_1 \wedge x_2 \wedge \bar{x}_3) \vee (x_1 \wedge \bar{x}_2 \wedge x_3)$

[6]

BITS PILANI, DUBAI CAMPUS
Dubai International Academic City, Dubai.
Third Year - II Semester 2016-17
Design and Analysis of Algorithms (CS F364)
Test -1 (Closed Book)

Date: 1.03.2017

Max Marks: 40

Time: 8.30 to 9.20 am, 50 minutes

Weightage: 20%

1. Consider modification of Binary search algorithm. The new algorithm splits the input array into three sets of approximately equal sizes and carry out the search recursively. Perform the time complexity analysis. Write the recurrence relation $T(n)$ for new algorithm and derive the general form of $T(n)$ up to i^{th} recurrence.

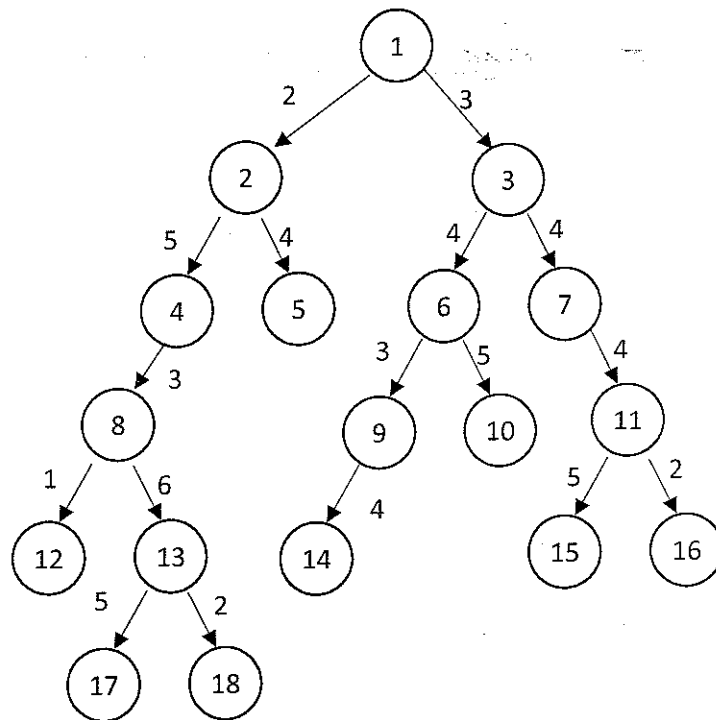
[8]

2. Write the complete algorithm for Merge-Sort and Merge.

[10]

3. For given following tree perform the vertex splitting. Consider $\delta = 7$. Show at least three vertex splits on entire tree. Also show final version of the tree after performing all vertex splitting.

[12]



4. Consider a scenario where there are n objects are given and user has a bag of capacity m . Each object has weight w_i and loss l_i . When user picks up x_i fraction of object i , loss incurred is $x_i \times l_i$. Objective is to fill the entire bag with minimum loss. Provide equation to minimize the loss and condition for feasible solution. For given $n=4$, $m=15$, $(w_1, w_2, w_3, w_4)=(6, 9, 4, 8)$ and $(l_1, l_2, l_3, l_4)=(12, 22, 41, 13)$. Find the optimum sequence which minimizes the loss using greedy method.

[10]

BITS PILANI, DUBAI CAMPUS
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II Semester 2016-17

Design and Analysis of Algorithms (CS F364)

Date: 1.06.2017

Compre Exam

Max Marks: 80

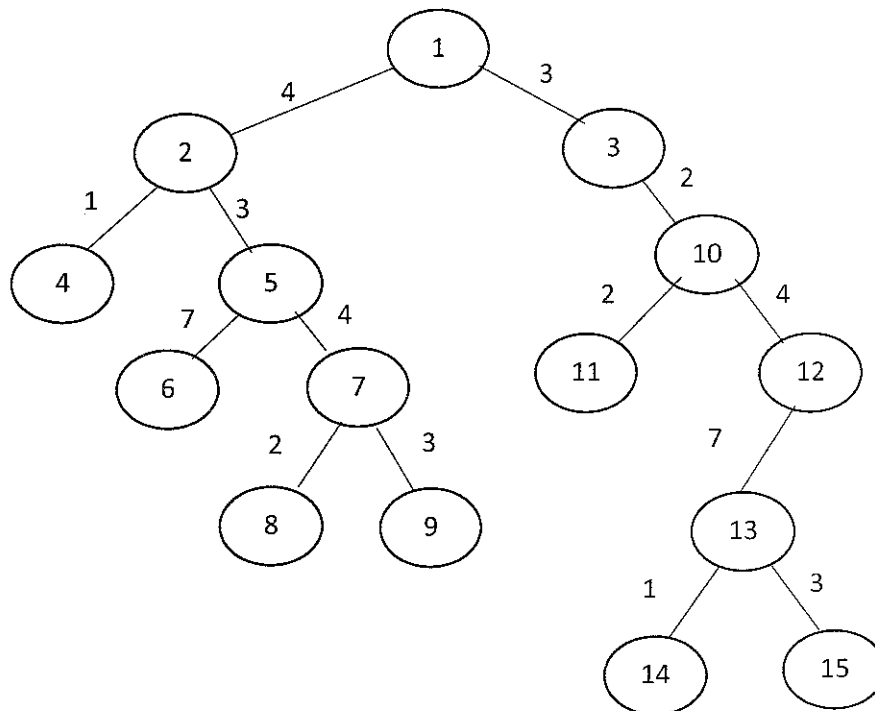
Time: 3 hours.

Weightage: 40%

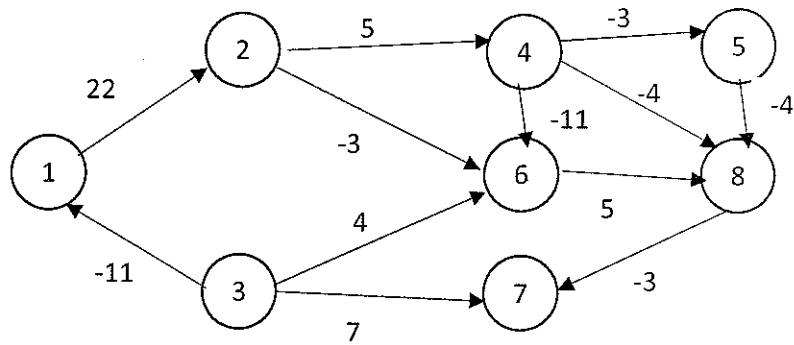
Note: Write Part A and Part B in separate answer paper

PART A (4X10=40 marks)

1. Write the general algorithm for Greedy method? Explain the Knapsack problem with all equations. For given number of objects $n=4$, $m=35$ and Profit $P=(22.4, 18.8, 10.35, 6.6)$ and weights $w=(12, 8, 13, 9)$. Write four feasible solutions and the optimum solution using Greedy method, where $0 \leq x_i \leq 1$.
- 2.A. Explain the Quick sort and write algorithm for Partition, Interchange and Quicksort.
2. B. Perform an iteration of Partition algorithm on $(38, 81, 22, 48, 13, 69, 93, 14, 45, 58, 79, 72)$ with $v=38$ as pivot element. Show at each step, values of variables of Partition algorithm.
3. For given tree, perform the tree vertex splitting using TVS algorithm, when
i) $\delta = 4$ and ii) $\delta = 7$.



4. Write the BellmanFord Algorithm to find single source shortest Paths. For given graph determine minimum distance matrix for single source/ all destination shortest path.



PART B(4X10=40 marks)

1. Describe CNF-Satisfiability and Clique decision problems. Write the proof to show that CNF-Satisfiability reduces to Clique decision problems and vice versa. Provide an example to show that a graph can be formed by CNF.

2. Given a list of items, we say that Y is between X and Z if either [X precedes Y and Y precedes Z] or [Z precedes Y and Y precedes X]. Thus, for example, in the list [C,A,D,E,B,F], D is between C and B; E is between F and D; and so on. This is called as Betweenness Problem. Write a non-deterministic algorithm to perform Betweenness Problem. Write what are input to the non-deterministic algorithm and suitably verify whether non-deterministic algorithm is success or failure.

3. Consider we have given bin packing problem with capacity of bins $L=15$, $n=7$ $(11,12...17)=(10,1,5,7,6,8,2)$. Perform the bin packing using

i. First fit ii. Best fit iii. First fit decreasing FFD iv. Best fit decreasing BFD

v. Best fit increasing order. (In best fit increasing order method, arrange the items from 11 to 17 in increasing order and then perform Best fit bin packing on them.)

In each method, depict the bin packing using diagram and how much left unfilled.

4 A. Describe what is a flow network, capacity condition and conservation condition.

4 B. Consider following flow network. A path is given as list of edges: $(a,d)(d,c)(c,f)(f,i)(i,h)(h,j)$ and the flow of this path is 10 units. Draw the graph showing the residual graph G_f . Find an augmented path in the residual graph G_f with flow of 8 units. Write the augmented path in terms of list of edges.

